

PELKOSENNIEMI PYHATUNTURI, Finland

WMO: 027050

Lat: 67.022N

Lon: 27.218E

Elev: 488

StdP: 95.60

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.4	-19.6	-24.0	0.5	-21.2	-22.0	0.6	-19.1	23.8	-8.0	22.2	-7.6	8.7	180	1.086

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	23.4	15.3	21.1	14.3	19.3	13.3	16.8	20.9	15.5	19.9	14.3	17.9	6.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	11.4	18.3	13.8	10.4	16.6	12.7	9.7	15.4	49.0	21.6	44.9	20.1	41.6	17.9	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.7	16.9	DB	-24.2	24.5	2.0	2.8	-25.6	26.5	-26.8	28.1	-27.9	29.7	-29.3	31.7	(4)
(5)			WB	-23.4	18.1	2.9	1.6	-25.5	19.2	-27.2	20.2	-28.8	21.1	-30.9	22.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-10.9	-9.8	-6.9	-1.5	5.0	9.7	13.7	11.5	6.9	-0.7	-4.9	-7.5	(6)
(7)		DBStd	9.33	4.72	5.14	3.81	3.66	5.20	4.60	4.31	3.64	3.46	3.59	3.67	4.21	(7)
(8)		HDD10.0	3767	647	555	523	346	173	59	14	24	103	332	448	542	(8)
(9)		HDD18.3	6546	906	788	782	596	413	261	154	213	344	591	698	801	(9)
(10)		CDD10.0	278	0	0	0	0	18	51	129	71	9	0	0	0	(10)
(11)		CDD18.3	16	0	0	0	0	1	3	11	2	0	0	0	0	(11)
(12)		CDH23.3	60	0	0	0	0	2	8	41	9	0	0	0	0	(12)
(13)		CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.8	9.9	10.3	10.4	8.3	8.0	7.5	6.7	7.5	8.4	8.6	9.8	10.5	(14)
(15)	Precipitation	PrecAvg	564	38	34	30	32	43	62	75	61	53	50	46	41	(15)
(16)		PrecMax	736	72	74	66	80	87	101	116	115	110	101	70	70	(16)
(17)		PrecMin	430	16	13	6	10	13	16	20	19	7	15	14	11	(17)
(18)		PrecStd	78	14	16	15	16	17	24	25	25	26	21	17	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	1.6	2.3	12.0	22.2	24.6	26.7	24.8	15.8	8.0	4.2	0.6	(19)
(20)			MCWB	-0.4	0.7	0.5	6.3	12.0	15.5	17.2	17.6	11.7	7.5	3.7	0.2	(20)
(21)		2%	DB	-1.7	-0.6	0.5	7.7	19.2	21.4	24.2	21.0	14.1	6.3	2.3	-0.5	(21)
(22)			MCWB	-2.0	-0.9	-0.6	4.0	11.2	14.3	15.7	14.7	10.8	5.4	1.9	-0.8	(22)
(23)		5%	DB	-3.0	-1.5	-0.8	5.0	15.9	18.8	22.1	18.9	12.8	5.0	0.8	-1.2	(23)
(24)			MCWB	-3.2	-1.7	-1.4	1.9	9.6	12.4	14.8	13.6	10.2	4.3	0.7	-1.5	(24)
(25)		10%	DB	-4.4	-2.9	-1.7	3.2	12.8	16.5	20.1	16.6	11.4	4.0	-0.2	-2.0	(25)
(26)			MCWB	-4.6	-3.1	-2.2	0.8	7.9	11.1	14.0	12.3	9.5	3.5	-0.4	-2.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	1.0	0.9	6.8	12.7	17.0	18.7	18.2	13.7	7.9	3.9	0.3	(27)
(28)			MCDWB	0.1	1.6	1.8	11.2	19.3	21.9	22.6	23.4	14.9	7.9	4.3	0.4	(28)
(29)		2%	WB	-2.0	-0.9	-0.4	4.3	11.7	15.4	17.2	16.0	11.7	5.6	1.9	-0.8	(29)
(30)			MCDWB	-1.6	-0.7	0.2	7.5	18.1	20.8	20.8	19.5	13.0	5.9	2.2	-0.6	(30)
(31)		5%	WB	-3.1	-1.7	-1.3	2.3	10.3	13.6	16.1	14.3	10.8	4.5	0.4	-1.4	(31)
(32)			MCDWB	-2.9	-1.5	-0.8	4.5	15.6	17.1	20.4	17.1	12.2	4.9	0.6	-1.3	(32)
(33)		10%	WB	-4.6	-3.0	-2.1	1.1	8.4	11.9	15.1	13.3	9.9	3.5	-0.5	-2.3	(33)
(34)			MCDWB	-4.5	-2.9	-1.8	2.8	12.5	15.1	19.4	15.8	10.9	3.9	-0.4	-2.1	(34)
(35)	Mean Daily Temperature Range	MDBR	4.3	4.0	4.3	4.9	6.6	6.5	6.5	5.5	4.2	3.1	3.4	3.6	(35)	
(36)		5% DB	MCDBR	5.6	4.7	4.6	6.5	10.0	8.9	8.3	7.6	5.7	4.1	4.1	3.9	(36)
(37)			MCWBR	5.6	4.5	4.0	3.9	5.3	4.4	3.8	3.9	3.8	3.6	4.4	3.9	(37)
(38)		5% WB	MCDWB	5.6	4.6	4.1	6.0	9.5	8.2	7.6	6.3	5.0	4.0	4.1	3.9	(38)
(39)			MCWBR	5.6	4.4	3.8	3.9	5.5	4.7	4.0	3.9	3.7	3.6	4.1	4.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.180	0.245	0.278	0.309	0.324	0.326	0.338	0.321	0.298	0.267	0.348	N/A	(40)	
(41)		taud	1.923	2.118	2.198	2.254	2.471	2.505	2.483	2.508	2.555	2.355	2.777	N/A	(41)	
(42)		Ebn at Noon	192	593	774	848	878	884	858	838	777	582	223	0	(42)	
(43)		Edh at Noon	13	50	80	100	91	90	89	78	59	40	15	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.06	0.56	2.05	3.87	4.79	5.10	4.70	3.43	1.78	0.60	0.12	0.00	(44)	
(45)		RadStd	0.00	0.09	0.19	0.31	0.42	0.40	0.47	0.32	0.17	0.12	0.02	0.00	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (34 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page